
Splunk SIEM Lab

Real Attack Detection & Security Monitoring

Field	Details
Lab Title	Splunk SIEM Lab with Real Attack Detection
Category	Security Information & Event Management (SIEM)
Platform	VMware Workstation 25H2u1 + Splunk Enterprise
Tools Used	Splunk, Kali Linux, Ubuntu Server, Apache2, Splunk UF

Objective

The primary objective of this lab is to build a functional Security Information and Event Management (SIEM) environment using Splunk Enterprise. This lab simulates a real-world attack scenario where a web server is exposed to potential threats, with all activity forwarded to a Splunk server for real-time monitoring, indexing, and analysis.

By the end of this lab, you will be able to:

- Deploy and configure Splunk Enterprise as a centralized log management server
- Install and configure the Splunk Universal Forwarder on a target Linux machine
- Simulate brute-force SSH attacks and web-based attacks using Kali Linux
- Write SPL (Splunk Processing Language) queries to detect and analyze attacks
- Build Splunk dashboards that visualize security events in real time
- Demonstrate understanding of SIEM architecture and log correlation

Requirements

Hardware Requirements

Minimum specs for running 3 VMs simultaneously:

- **RAM:** Minimum 12 GB (16 GB recommended)
- **CPU:** 4+ cores (Intel VT-x or AMD-V enabled in BIOS)
- **Storage:** At least 80 GB free disk space
- **Host OS:** Windows 10/11 or Linux

Software Requirements

Software	Version / Details	Download URL
VMware Workstation Pro	25H2u1 (v25219725)	https://support.broadcom.com
Kali Linux ISO	2024.x (64-bit)	https://www.kali.org/get-kali/
Ubuntu Server ISO	22.04 LTS or 24.04 LTS	https://ubuntu.com/download/server
Splunk Enterprise	Latest (Free Trial / Dev)	https://www.splunk.com/en_us/download/splunk-enterprise.html
Splunk Universal Forwarder	Match Splunk version	https://www.splunk.com/en_us/download/universal-forwarder.html

VM Configuration Summary

VM Name	OS	RAM	Role	IP (Example)
Splunk-Server	Ubuntu 22.04	4 GB	SIEM / Log Collector	192.168.1.10
Web-Server	Ubuntu 22.04	2 GB	Target (Apache2 + UF)	192.168.1.20
Kali-Attacker	Kali Linux	2 GB	Attacker Machine	192.168.1.30

Theory

What is SIEM?

Security Information and Event Management (SIEM) is a cybersecurity technology that provides real-time analysis of security alerts generated by applications, network hardware, and systems. A SIEM aggregates log data produced throughout an organization's infrastructure, from host systems and applications to network and security devices, and performs correlation, alerting, and reporting.

What is Splunk?

Splunk is a leading SIEM and data analytics platform that indexes machine-generated data from any source — logs, events, metrics, and traces. It allows security teams to search, monitor, analyze, and visualize data in real time using SPL (Splunk Processing Language). Splunk is widely used in SOC (Security Operations Center) environments for threat detection, incident response, and compliance reporting.

Key Components in This Lab

- **Splunk Enterprise:** The central SIEM server — receives, indexes, and allows analysis of all logs.
- **Splunk Universal Forwarder (UF):** A lightweight agent installed on the web server that forwards log files to the Splunk server.
- **Apache2 Web Server:** Generates access and error logs that simulate real-world web traffic and attack attempts.
- **Kali Linux:** Used as the attacker machine to simulate brute-force SSH attacks and web scanning.
- **SPL (Splunk Processing Language):** A query language used to search and analyze indexed data in Splunk.

Attack Vectors Simulated

- SSH Brute Force Attack (using Hydra or Medusa from Kali Linux)
- Web Directory Scanning (using Nikto or Gobuster)
- Failed Login Monitoring via /var/log/auth.log
- Apache Access Log Analysis for suspicious IP patterns

Lab Setup Steps

Step 1 — Install VMware Workstation

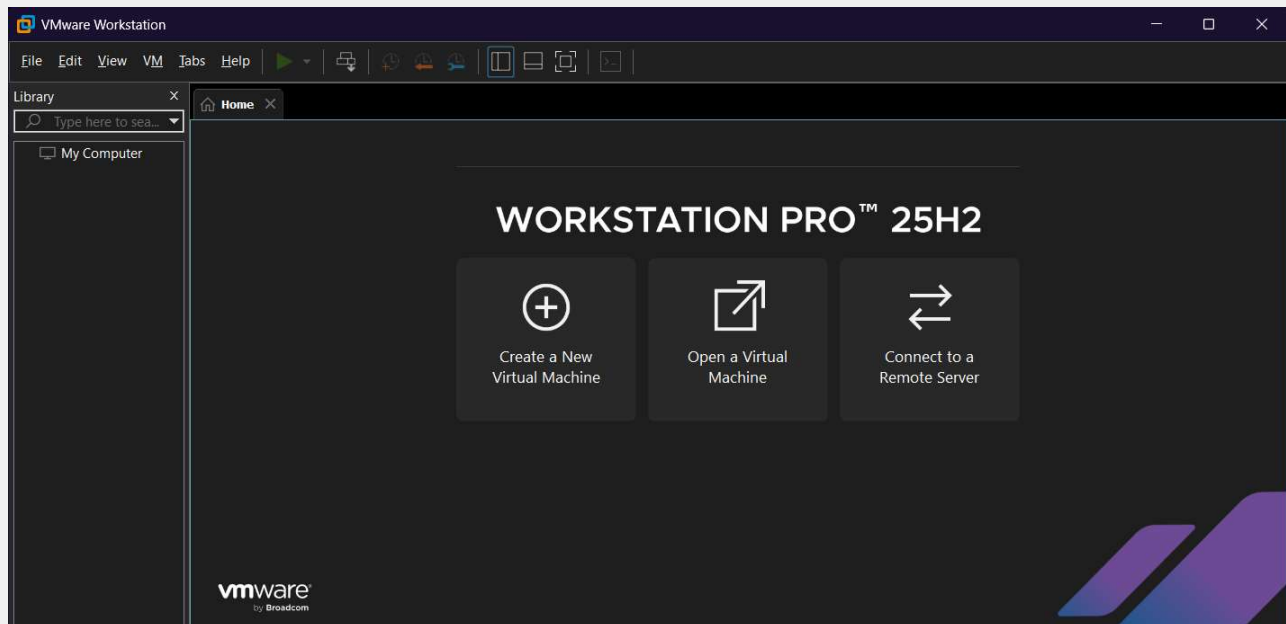
Download VMware Workstation Pro 25H2u1 from Broadcom's support portal:

→

<https://support.broadcom.com/group/ecx/productdownloads?subfamily=VMware+Workstation+Pro>

Run the installer: VMware-Workstation-Full-25H2u1-25219725.exe

- Accept the license agreement and install with default settings
- Reboot after installation if prompted
- Enable virtualization in BIOS if VMs fail to start (Intel VT-x / AMD-V)



Screenshot: VMware Workstation Pro installed and running

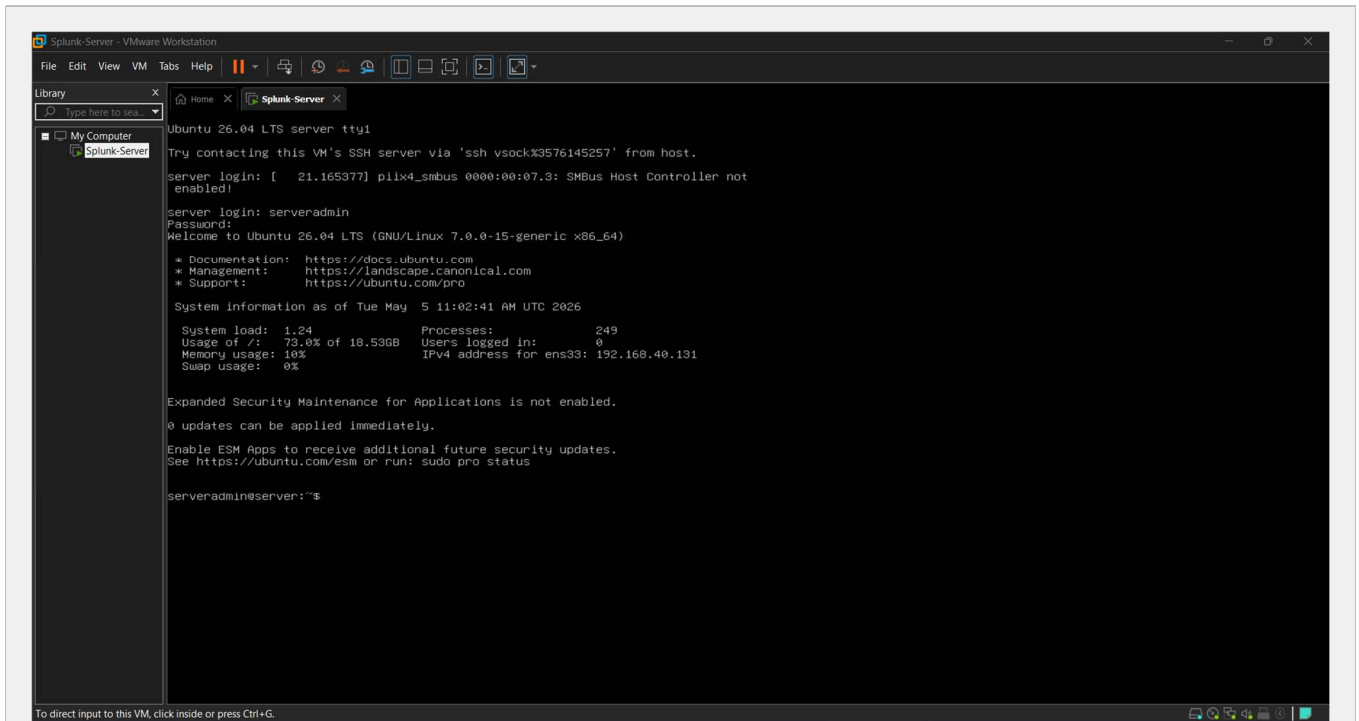
Step 2 — Create VM 1: Splunk Server (Ubuntu 22.04)

Download Ubuntu Server 22.04 LTS ISO:

→ <https://ubuntu.com/download/server>

1. Open VMware → Click 'Create a New Virtual Machine'
2. Select 'Typical' → Browse and select the Ubuntu ISO
3. Name: Splunk-Server | Location: your preferred path

4. Disk Size: 40 GB (Split into multiple files)
5. Customize Hardware → RAM: 4096 MB, CPU: 2 cores
6. Network: Host-Only or NAT (consistent across all VMs)
7. Complete installation. Set username: splunkadmin



Screenshot: Splunk-Server VM creation in VMware

Step 3 — Install Splunk Enterprise on Splunk-Server

Download Splunk Enterprise (.deb) from:

→ https://www.splunk.com/en_us/download/splunk-enterprise.html

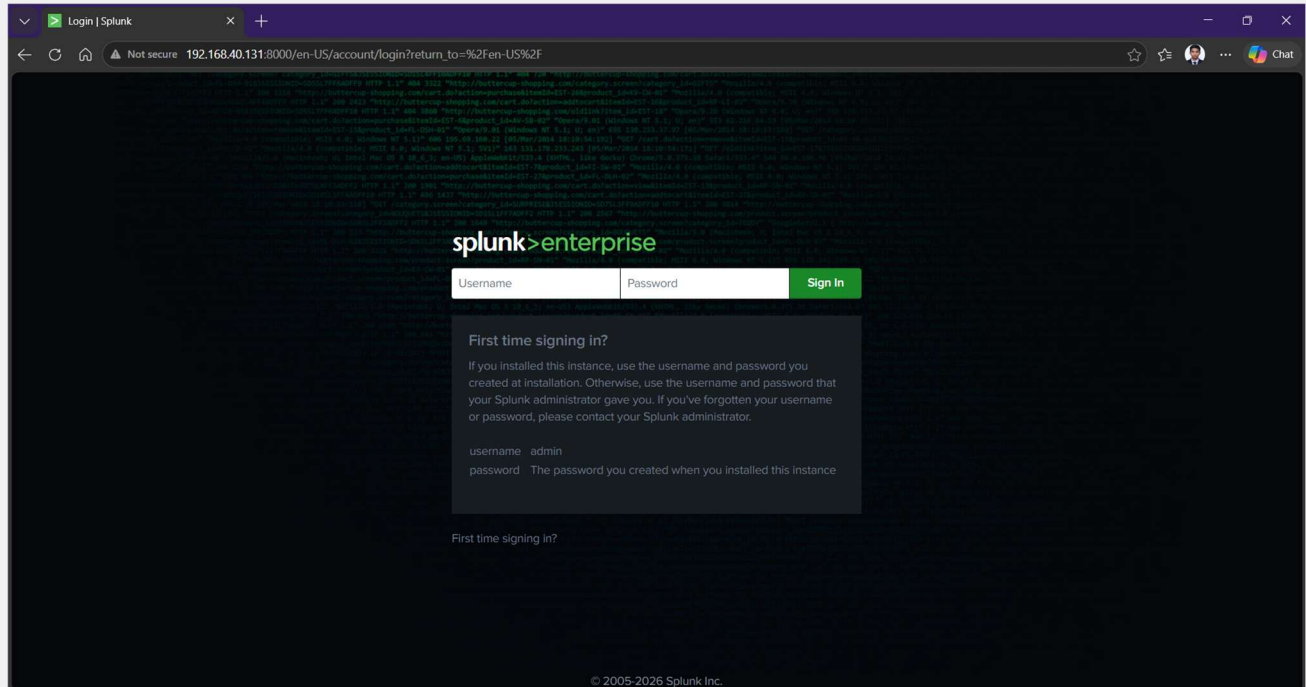
Transfer the .deb file to the VM and run:

```
sudo dpkg -i splunk-<version>-linux-2.6-amd64.deb
sudo /opt/splunk/bin/splunk start --accept-license
sudo /opt/splunk/bin/splunk enable boot-start
```

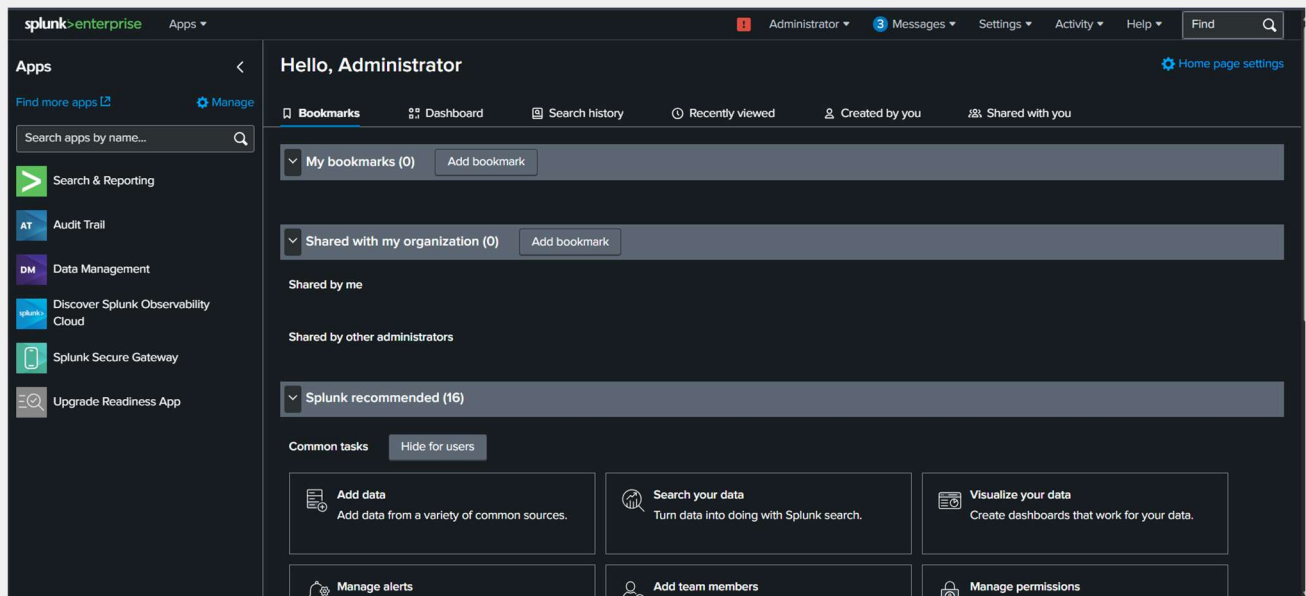
Access the Splunk web UI in your host browser:

→ <http://<Splunk-Server-IP>:8000>

Login with username: admin | Password: (set during install)



Screenshot: Splunk Web UI login page at port 8000



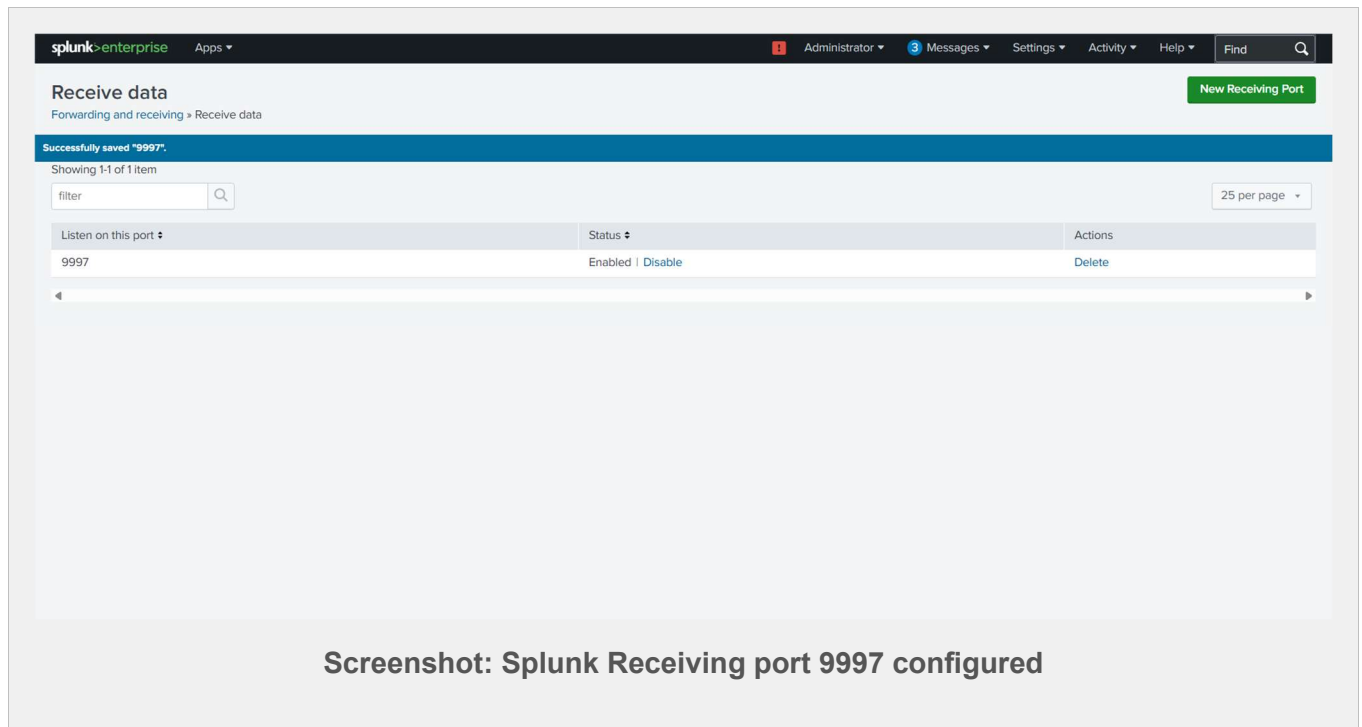
Screenshot: Splunk Dashboard after first login

Step 4 — Enable Splunk Receiving Port 9997

On the Splunk Server, enable the receiver to accept forwarded logs:

```
| sudo /opt/splunk/bin/splunk enable listen 9997 -auth admin:<password>
```

Or via Web UI: Settings → Forwarding and Receiving → Configure Receiving → Add port 9997



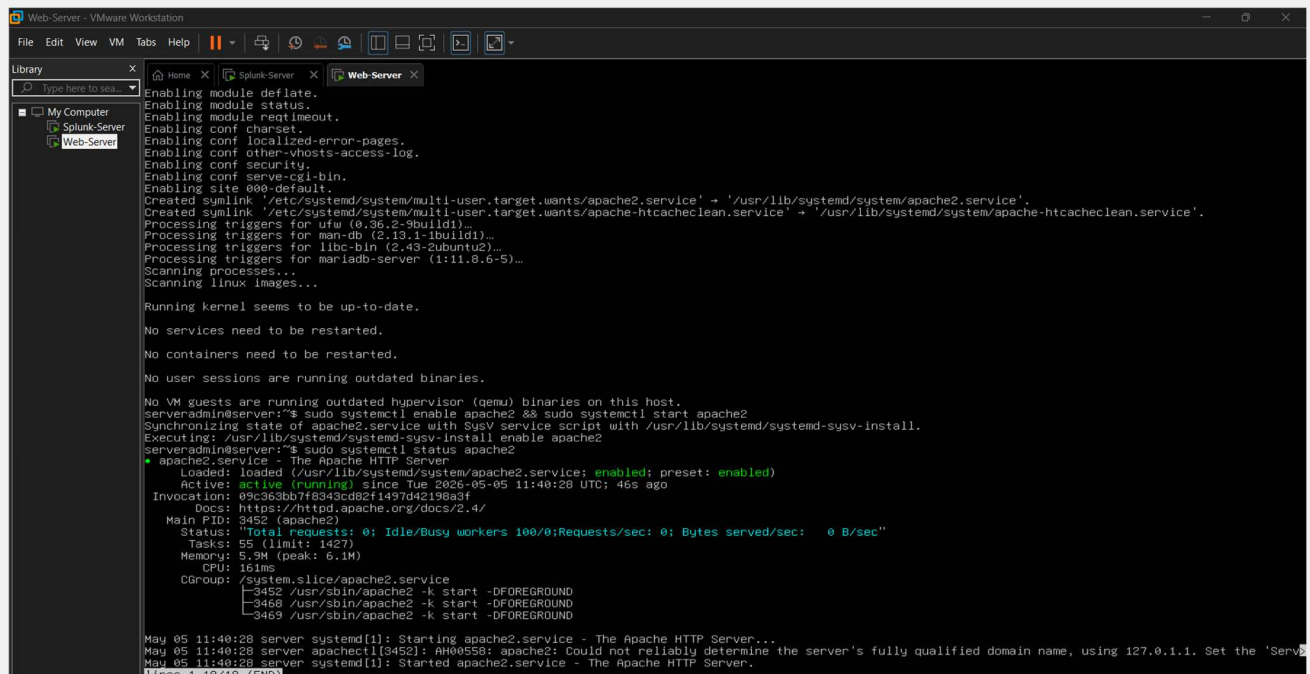
Step 5 — Create VM 2: Web Server (Ubuntu 22.04)

Use a second Ubuntu 22.04 VM (2 GB RAM, 20 GB disk). Install the LAMP stack:

```
| sudo apt update && sudo apt install apache2 mariadb-server -y  
| sudo systemctl enable apache2 && sudo systemctl start apache2
```

Verify Apache is running:

```
| sudo systemctl status apache2
```



```
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/system/apache2.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' -> '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2)...
Processing triggers for mariadb-server (1:11.8.6-5)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
serveradmin$server:~$ sudo systemctl enable apache2 && sudo systemctl start apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
serveradmin$server:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-05 11:48:28 UTC; 46s ago
   Invocation: 09c363bb7f6343cd82f1497d42198a3f
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3452 (apache2)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 55 (limit: 1427)
    Memory: 5.9M (peak: 6.1M)
       CPU: 161ms
    CGroup: /system.slice/apache2.service
            └─3452 /usr/sbin/apache2 -k start -DFOREGROUND
              └─3469 /usr/sbin/apache2 -k start -DFOREGROUND
              └─3469 /usr/sbin/apache2 -k start -DFOREGROUND

May 05 11:48:28 server systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 05 11:48:28 server apache2[3452]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive dynamically to determine the server name.
May 05 11:48:28 server systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Screenshot: Apache2 running on Web-Server VM

Step 6 — Install Splunk Universal Forwarder on Web-Server

Download the Universal Forwarder (matching version) from:

→ https://www.splunk.com/en_us/download/universal-forwarder.html

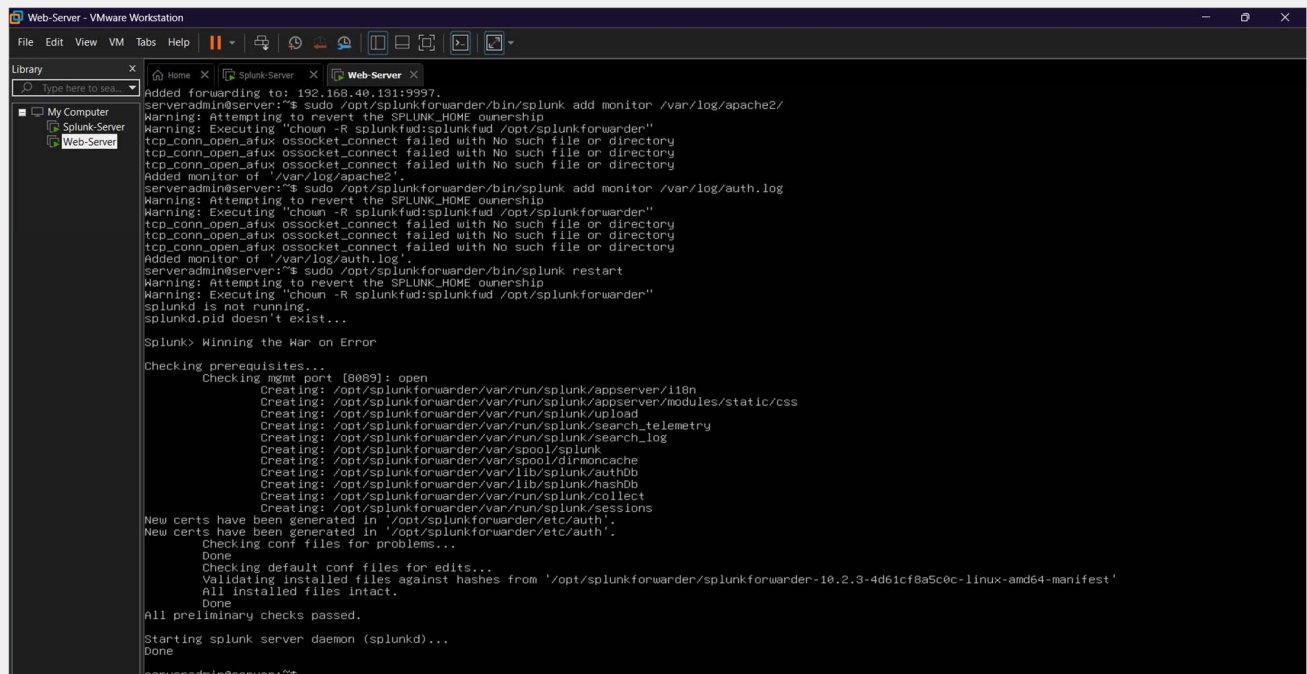
```
sudo dpkg -i splunkforwarder-<version>-linux-2.6-amd64.deb
sudo /opt/splunkforwarder/bin/splunk start --accept-license
sudo /opt/splunkforwarder/bin/splunk enable boot-start
```

Add the Splunk server as the forward destination:

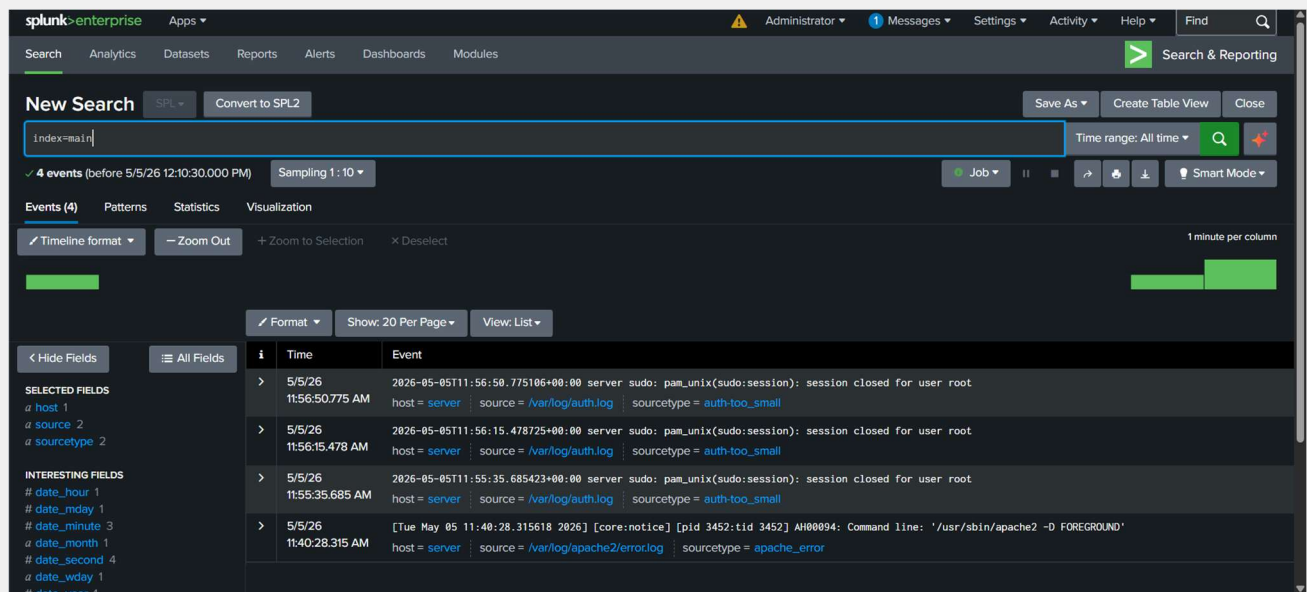
```
sudo /opt/splunkforwarder/bin/splunk add forward-server <Splunk-Server-IP>:9997
```

Add directories to monitor (logs to forward):

```
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/apache2/
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log
sudo /opt/splunkforwarder/bin/splunk restart
```



Screenshot: Universal Forwarder configured and running on Web-Server



Screenshot: Logs appearing in Splunk from Web-Server (index=main)

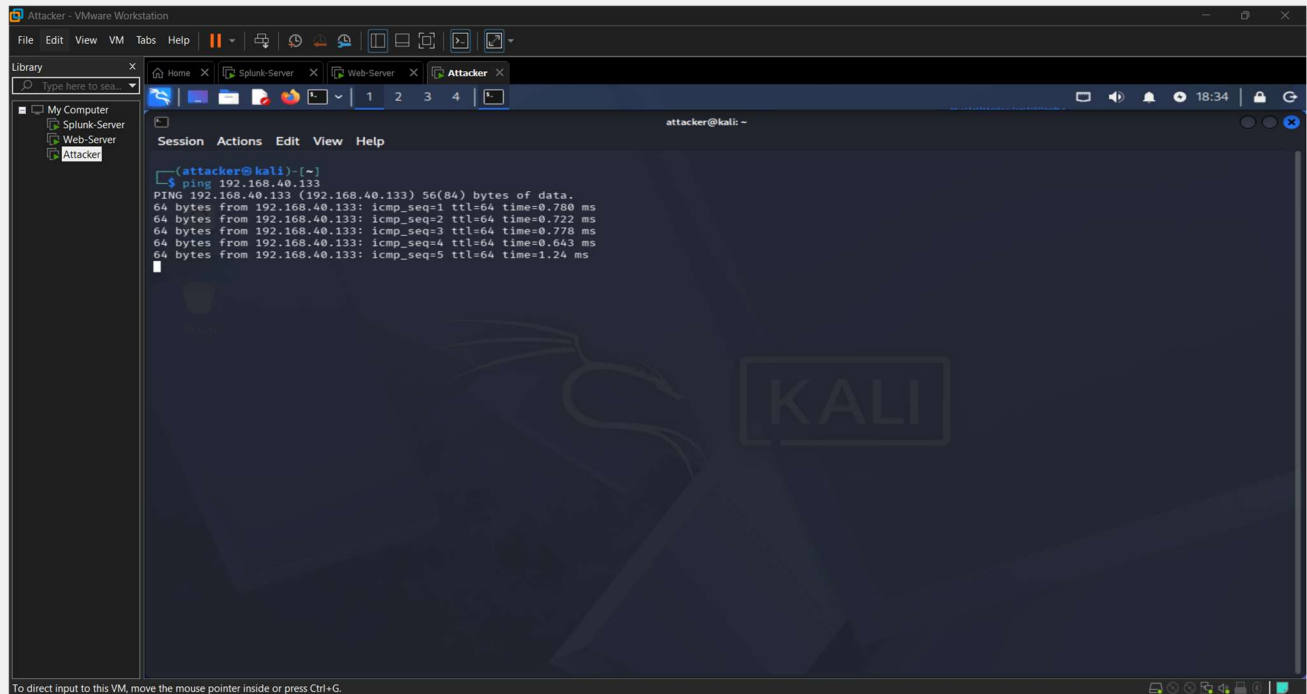
Step 7 — Create VM 3: Kali Linux (Attacker)

Download Kali Linux ISO:

→ <https://www.kali.org/get-kali/#kali-installer-images>

Create a VM (2 GB RAM, 20 GB disk) and install Kali Linux. Ensure it is on the same network as the other VMs. Verify connectivity:

```
ping <Web-Server-IP>
```



Screenshot: Kali Linux VM running, ping to Web-Server successful

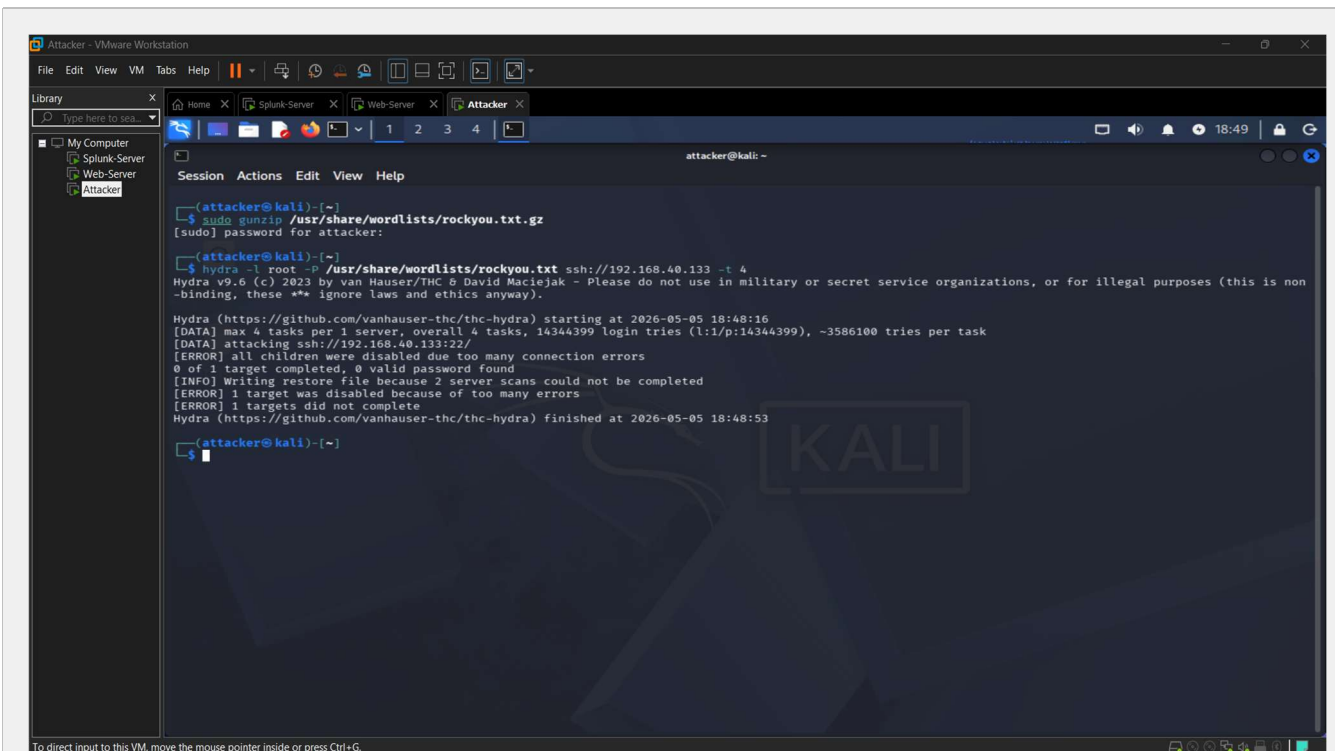
Lab Steps — What We Did

Phase 1: Simulate SSH Brute Force Attack

From Kali Linux, run an SSH brute-force attack against the Web-Server using Hydra:

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://<Web-Server-IP> -t 4
```

This generates multiple failed SSH login attempts in /var/log/auth.log, which the Splunk Universal Forwarder forwards to Splunk in real time.



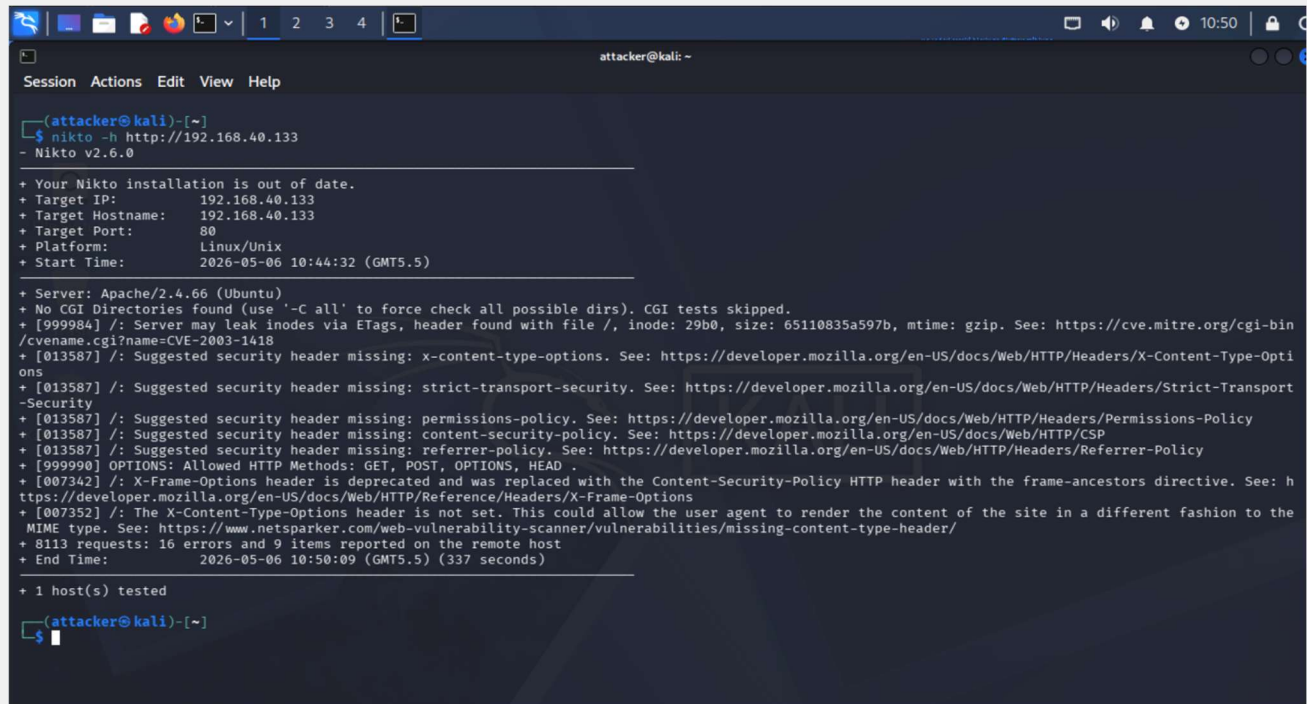
Screenshot: Hydra brute force command running on Kali Linux

Phase 2: Simulate Web Directory Scanning

From Kali, run a web scan against the Apache web server using Nikto:

```
nikto -h http://<Web-Server-IP>
```

These generate entries in /var/log/apache2/access.log and error.log.



```
attacker@kali: ~  
Session Actions Edit View Help  
  
(attacker@kali)-[~]  
$ nikto -h http://192.168.40.133  
- Nikto v2.6.0  
  
+ Your Nikto installation is out of date.  
+ Target IP: 192.168.40.133  
+ Target Hostname: 192.168.40.133  
+ Target Port: 80  
+ Platform: Linux/Unix  
+ Start Time: 2026-05-06 10:44:32 (GMT5.5)  
  
+ Server: Apache/2.4.66 (Ubuntu)  
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.  
+ [999984] /: Server may leak inodes via ETags, header found with file /, inode: 29b0, size: 65110835a597b, mtime: gzip. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418  
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options  
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security  
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy  
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP  
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy  
+ [999990] OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD .  
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Options  
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/  
+ 8113 requests: 16 errors and 9 items reported on the remote host  
+ End Time: 2026-05-06 10:50:09 (GMT5.5) (337 seconds)  
  
+ 1 host(s) tested  
  
(attacker@kali)-[~]  
$
```

Screenshot: Nikto scan running from Kali against Web-Server

Phase 3: Search and Detect Attacks in Splunk

Go to Splunk Web UI → Search & Reporting. Run the following SPL queries:

Detect Failed SSH Logins (Brute Force Detection):

Note: Splunk does not automatically extract the source IP from auth.log. We need to first create a custom field extraction for `src_ip` using Splunk's built-in Field Extractor tool.

Steps to Add `src_ip` Field Extraction in Splunk:

- In Splunk Search, run: `index=main source="/var/log/auth.log" "Failed password"`
- In the left sidebar, scroll down and click “+ Extract New Fields”
- A sample log event appears. Click on the attacker IP address (e.g. 192.168.40.134) in the raw log to highlight it
- A box appears asking “What is this field called?” — type: `src_ip` then click **Add Extraction**
- Splunk generates a regex preview — verify it looks correct, then click **Save**
- The field `src_ip` now appears in the left sidebar under **Interesting Fields** for all future searches automatically ✓

The screenshot shows the Splunk Field Extractor interface. On the left, the 'INTERESTING FIELDS' list includes # date_hour 1, # date_minute 1, # date_month 1, # date_second 26, # date_wday 1, # date_year 1, # date_zone 1, # index 1, # linecount 1, # punct 1, # splunk_server 1, # src_ip 1, # timeendpos 1, and # timestartpos 1. The main table displays event data with columns for Time and Event. The 'src_ip' field is highlighted in the 'Values' section, showing a count of 48 for the value 192.168.40.134.

Time	Event
5/5/26 1:18:53.250 PM	2026-05-05T13:18:53.250603+00:00 server sshd-session[5253]: Failed password for root from 192.168.40.134 port 49508 ssh2
5/5/26 1:18:52.845 PM	2026-05-05T13:18:52.845042+00:00 server sshd-session[5262]: Failed password for root from 192.168.40.134 port 49528 ssh2
5/5/26 1:18:51.213 PM	2026-05-05T13:18:51.213977+00:00 server sshd-session[5253]: Failed password for root from 192.168.40.134 port 49508 ssh2
5/5/26 1:18:50.967 PM	2026-05-05T13:18:50.967023+00:00 server sshd-session[5262]: Failed password for root from 192.168.40.134 port 49528 ssh2
5/5/26 1:18:48.418 PM	2026-05-05T13:18:48.418194+00:00 server sshd-session[5251]: Failed password for root from 192.168.40.134 port 49500 ssh2

Screenshot: src_ip field extraction saved in Splunk Field Extractor

Detect Failed SSH Logins (Brute Force Detection):

```
index=main source="/var/log/auth.log" "Failed password" | stats count by src_ip | sort -count
```

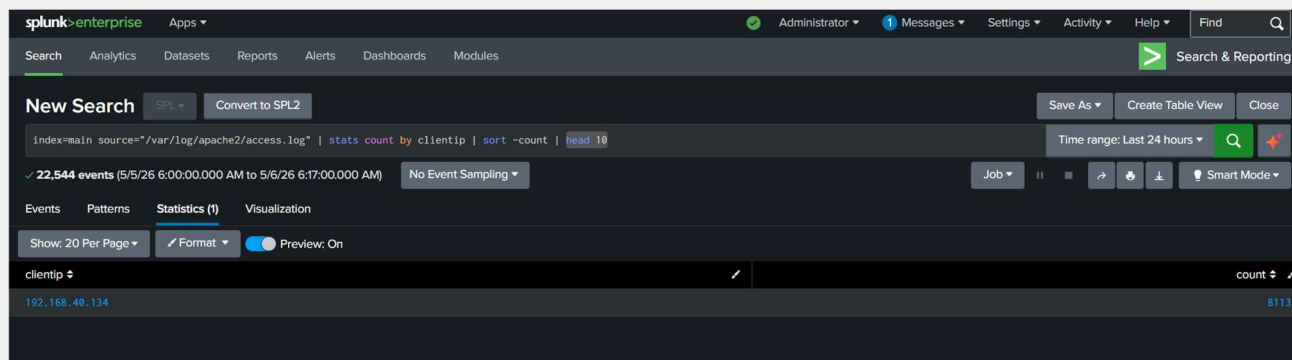
The screenshot shows the Splunk Search interface. The search bar contains the query: `index=main source="/var/log/auth.log" "Failed password" | stats count by src_ip | sort -count`. The results show 48 events. The 'Statistics (1)' tab is selected, displaying a table with the following data:

src_ip	count
192.168.40.134	48

Screenshot: SPL query results showing failed SSH login attempts by IP

Top Attacking IPs from Apache Logs:

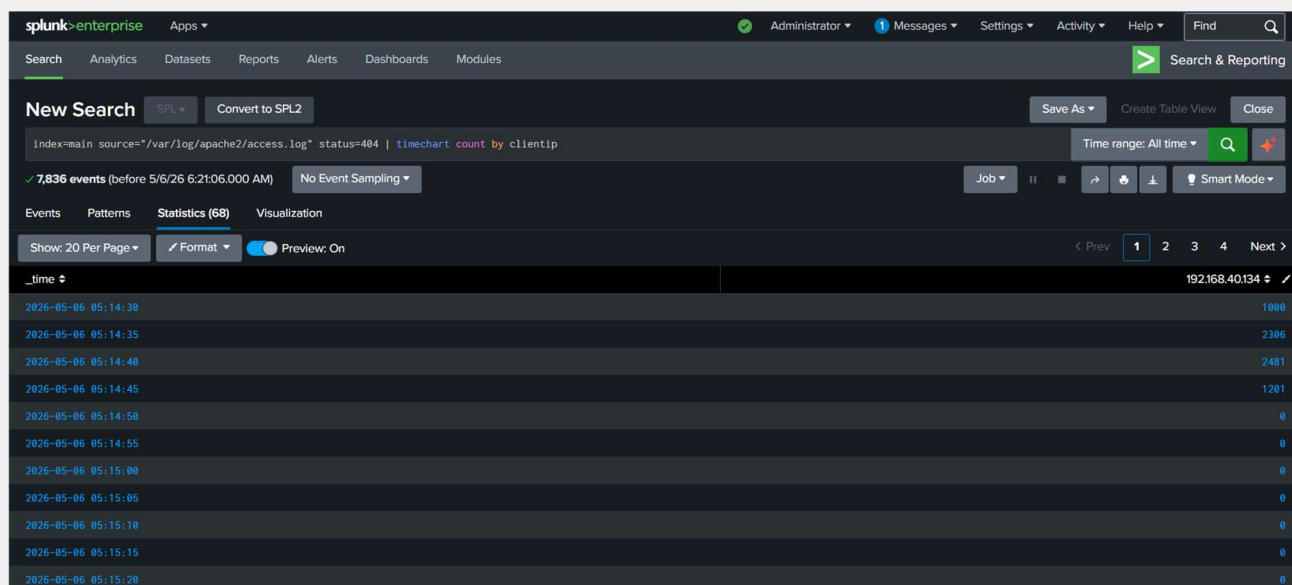
```
index=main source="/var/log/apache2/access.log" | stats count by clientip |  
sort -count | head 10
```



Screenshot: Top attacking IPs from Apache access logs in Splunk

Detect 404 Error Spikes (Directory Scanning):

```
index=main source="/var/log/apache2/access.log" status=404 | timechart count  
by clientip
```



Screenshot: 404 error spike timechart showing directory scan pattern

Real-time Event Feed:

```
index=main | head 50 | table _time, host, source, _raw
```

```
index=main | head 50 | table time, host, source, raw
```

New Search

SPL+ Convert to SPL2 Save As Create Table View Close

index=main | head 50 | table _time, host, source, _raw

Time range: All time [Search Icon] [Filter Icon]

✓ 50 events (before 5/6/26 6:25:00.000 AM) No Event Sampling Job ▾ || □ → ↺ ⬇️ Smart Mode ▽

Events Patterns Statistics (50) Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On < Prev 1 2 3 Next >

_time ↓	host ↑	source ↓	_raw ↓
2026-05-06 06:17:01.532	webserver	/var/log/auth.log	2026-05-06T06:17:01.532Z+00:00 webserver CRON[2812]: pam_unix(cron:session): session closed for user root
2026-05-06 06:17:01.513	webserver	/var/log/auth.log	2026-05-06T06:17:01.513Z+00:00 webserver CRON[2812]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-06 05:20:09.095	webserver	/var/log/apache2/error.log	[Wed May 06 05:20:09.095364 2026] [core:error] [pid 1617:tld 1667] [client 192.168.40.134:43250] AH!0244: invalid URI path (/./../././././././././././././././etc/shadow)
2026-05-06 05:20:09.000	webserver	/var/log/apache2/access.log	192.168.40.134 -- [06/May/2026:05:20:09 +0000] "GET /crossdomain.xml HTTP/1.1" 404 532 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 14_4) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.6248.19 Safari/537.36 Edg/141.0.6248.19"
2026-05-06 05:20:09.000	webserver	/var/log/apache2/access.log	192.168.40.134 -- [06/May/2026:05:20:09 +0000] "GET /clientaccesspolicy.xml HTTP/1.1" 404 532 "-" Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) EdgeIOS/142.0.6281.42 Mobile/ISE148 Safari/605.1.15"
2026-05-06 05:20:09.000	webserver	/var/log/apache2/access.log	192.168.40.134 -- [06/May/2026:05:20:09 +0000] "GET /api/v1/access-tokens HTTP/1.1" 404 532 "-" Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) EdgeIOS/142.0.6281.42 Mobile/ISE148 Safari/605.1.15"
2026-05-06	webserver	/var/log/apache2/access.log	192.168.40.134 -- [06/May/2026:05:20:09 +0000] "GET /my-site/_api/v1/access-tokens HTTP/1.1" 404 532 "-" Mozilla/5.0 (iPhone; CPU iPhone OS

Screenshot: Real-time event feed showing forwarded log events

Phase 4: Create Alerts in Splunk

In Splunk, save the brute force search as an alert:

1. Run the Failed SSH query above
2. Click 'Save As' → Alert
3. Name: SSH Brute Force Detected
4. Alert type: Real-time
5. Trigger condition: Number of results > 5
6. Add action: Log event / Send email
7. Save the alert

Save As Alert

Trigger Conditions

Trigger alert when: Number of Results

is greater than 0

in 1 minute(s)

Trigger: Once For each result

Throttle ☐

Trigger Actions

+ Add Actions

When triggered

Add to Triggered Alerts Remove

Severity Medium

Cancel Save

Screenshot: Splunk Alert configuration for SSH Brute Force

After saving the alert, the following SPL query was used to verify the brute force activity by counting failed SSH login attempts per source IP and filtering only IPs with more than 5 attempts:

```
index=main source="/var/log/auth.log" "Failed password" | stats count by src_ip | where count > 5 | sort -count
```

The query returned attacker IP 192.168.40.134 with **620 failed attempts**, confirming active brute force activity. This triggered the **SSH Brute Force Detected** alert which appeared in **Activity** → **Triggered Alerts**, confirming the alert is working correctly in real time.

Triggered Alerts

Filter

App Search & Report... Owner All owners Severity All severity Alert name All alerts

Showing 1 - 20 of 20 results Show: 20 per ... 1 of 1 pages

☐	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2026-05-06 07:58:25 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:20 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:17 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:16 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:11 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:10 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:05 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:58:04 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:59 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:58 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:53 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:50 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:47 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-05-06 07:57:46 UTC	SSH Brute Force Detected	search	Real-time	Medium	Digest	View Results Edit Alert Delete

Screenshot: SSH Brute Force Detected alert fired in Splunk Triggered Alerts

Phase 5: Build a Security Dashboard

In Splunk go to **Dashboards** → **Create New Dashboard**

- Title: Security Monitoring Dashboard
- Type: Classic Dashboards
- Permissions: Private
- Click **Create**

Panel 1 — Failed SSH Logins Over Time:

Click **Add Panel** → **New** → **Line Chart**

Search:

```
index=main source="/var/log/auth.log" "Failed password" | timechart count span=1m
```

Title: **Failed SSH Logins Over Time**



Screenshot: Failed SSH Logins Over Time timechart added to dashboard

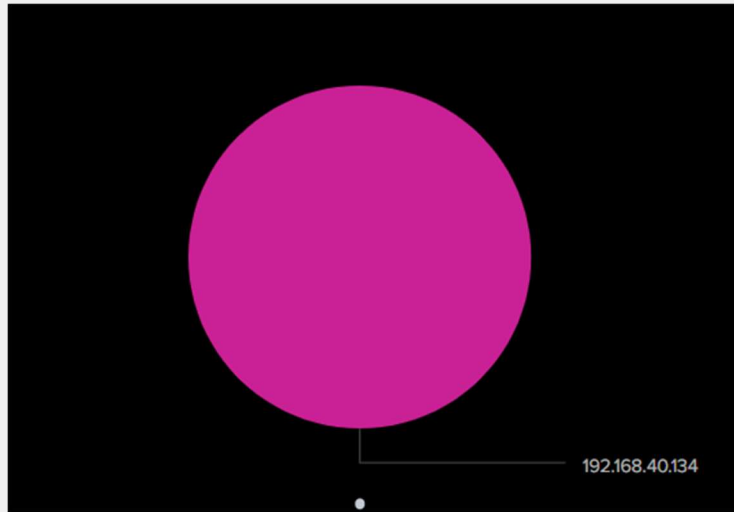
Panel 2 — Top Attacker IPs:

Click **Add Panel** → **New** → **Pie Chart**

Search:

```
index=main source="/var/log/auth.log" "Failed password" | stats count by src_ip | sort -count | head 10
```

Title: **Top Attacker Ips**



Screenshot: Top Attacker IPs pie chart added to dashboard

Panel 3 — 404 Error Spikes (Web Scanning):

Click **Add Panel** → **New** → **Column Chart**

Search:

```
index=main source="/var/log/apache2/access.log" status=404 | timechart count span=1m
```

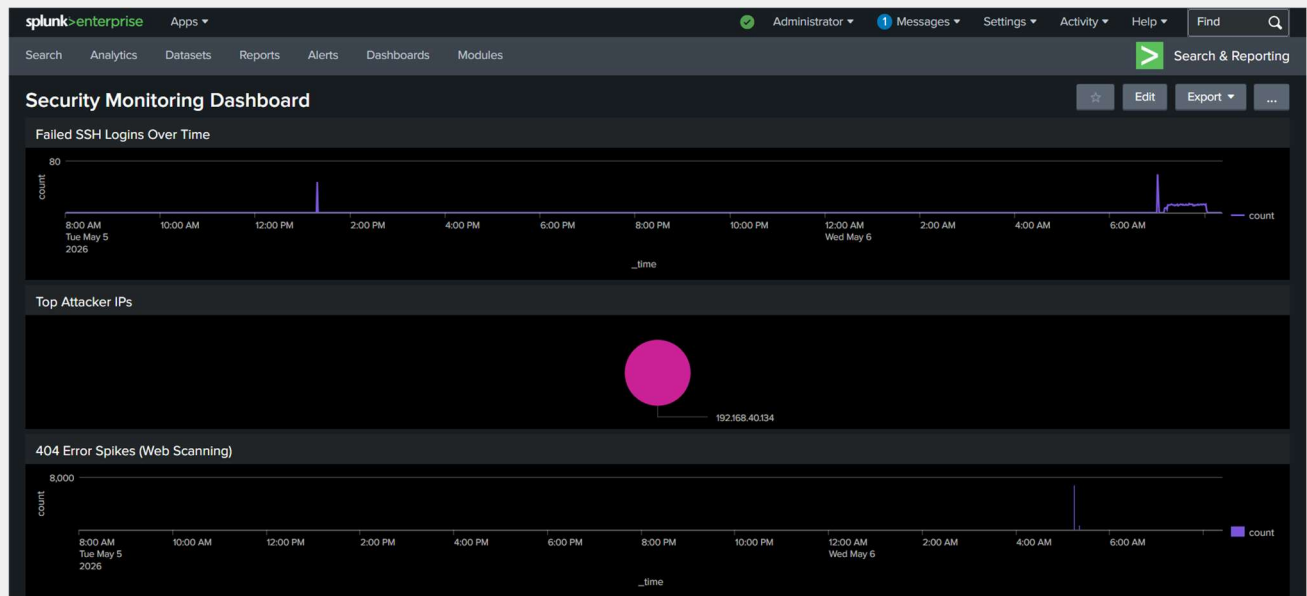
Title: 404 Error Spikes (Web Scanning)



Screenshot: 404 Error Spikes column chart added to dashboard

Save Dashboard:

Click **Save Dashboard** at the top right



Screenshot: Splunk Security Monitoring Dashboard with all panels

Conclusion

This Splunk SIEM Lab successfully demonstrated a complete end-to-end security monitoring workflow using industry-standard tools. By setting up a multi-VM environment with VMware Workstation, a dedicated Splunk SIEM server, a target Apache web server, and a Kali Linux attacker, we simulated real-world cyber attack scenarios and observed how they appear in log data.

Key takeaways from this lab:

- **SIEM Deployment:** Splunk Enterprise was configured from scratch as a log aggregation and analysis platform.
- **Log Forwarding:** The Splunk Universal Forwarder successfully shipped Apache and auth logs from the web server to the central SIEM in real time.
- **Attack Simulation:** SSH brute-force and web directory scanning attacks generated measurable, detectable log entries.
- **SPL Querying:** Custom SPL queries were written to detect attack patterns such as failed login spikes, 404 bursts, and IP anomalies.
- **Alerting:** Real-time Splunk alerts were configured to trigger automatically when attack thresholds were exceeded.
- **Dashboard:** A security dashboard was built to provide a single-pane-of-glass view of all security events.

This lab closely mirrors the responsibilities of a SOC Analyst and demonstrates skills that are directly applicable to roles in Cybersecurity, IT Operations, and Cloud Security. The experience gained — including working with Splunk Universal Forwarder, SPL, alerting, and dashboarding — aligns with the Splunk Core Certified User certification objectives.

References & Download URLs

- VMware Workstation Pro: <https://support.broadcom.com>
- Kali Linux: <https://www.kali.org/get-kali/>
- Ubuntu Server: <https://ubuntu.com/download/server>
- Splunk Enterprise: https://www.splunk.com/en_us/download/splunk-enterprise.html
- Splunk Universal Forwarder: https://www.splunk.com/en_us/download/universal-forwarder.html
- Splunk BOTS Dataset (for extra practice): <https://github.com/splunk/botsv3>
- Splunk SPL Reference: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference>